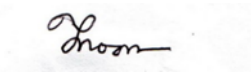
# CSE 4000: Final Year Design Project (FYDP)
## Reflective Journal Entry
## United International University (UIU)

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| **Name: Ibtida Ara** | | **Sec: J** |
| **Group: 262-074** | **Date: 23 September,2026** | **Week: 10** |

| **Reflective Entry (Weekly activities)** | **Signature and remarks of faculty** |
|---|---|
| • Conducted a team meeting and discussed next steps for the project.<br>• Modified the gap analysis table to incorporate findings from recently read papers.<br>• Read the paper titled "**Do Small Models Use the Law You Give Them? Context-Injected Fine-Tuning for Legal QA in Bangladesh**".<br>• Key Findings:<br>▪ Fine-tuning Qwen3.5 on context-injected RAG prompts produced significant accuracy gains at smaller scales under strict consistency scoring — 0.8B gained +14.1 mean strict accuracy points, 2B gained +7.8 points.<br>▪ Diminishing returns at larger scale: the 4B fine-tuned model showed no overall net benefit (-2.1 mean strict accuracy shift), as modest Bangla gains were offset by regressions in several English conditions.<br>▪ Near-total elimination of language drift: base models answered Bangla legal questions in English 44.0%–53.2% of the time; fine-tuning reduced this to 0.2%–0.7% across all scales (adjusted $p < .001$).<br>▪ Context utilization vs. retrieval quality: given identical retrieved context (BM25 or FAISS), fine-tuned smaller models extracted far more value from statutory text than base counterparts — showing retrieval quality alone is not the bottleneck in small legal LLMs. |  |